

An investigation of the internal structure of shock profiles for shock capturing schemes

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Abstract

The theoretical understanding of discrete shock transitions obtained by shock capturing schemes is very incomplete. Previous experimental studies indicate that discrete shock transitions obtained by shock capturing schemes can be modeled by continuous functions, so called continuum shock profiles. However, the previous papers have focused on linear methods. We have experimentally studied the trajectories of discrete shock profiles in phase space for a range of different high resolution shock capturing schemes, including Riemann solver based flux limiter methods, high resolution central schemes and ENO type methods. In some cases, no continuum profiles exists. However, in these cases the point values in the shock transitions remain bounded and appear to converge toward a stable limit cycle. The possibility of such behavior was anticipated in *Bultelle, Grassin and Serre*, 1998, but no specific examples, or other evidence, of this behavior have previously been given. In other cases, our results indicate that continuum shock profiles exist, but are very complicated. We also study phase space orbits with regard to post shock oscillations.

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1. Introduction

A major computational difficulty when solving hyperbolic conservation laws is to handle the discontinuities that can arise in the solution. One wants the numerical solution to be accurate with high resolution, avoiding oscillations or excessive smearing around shocks and contact discontinuities. There exists a broad range of so called high resolution shock capturing schemes of different types, see [19] and references therein. One type is Riemann solver based flux limiter methods, where the flux is a weighted average of a lower order flux and a higher order flux. The weighting depends nonlinearly on the solution. Another type is high resolution central schemes. These methods involve no Riemann solver. Instead, they are based on the Lax–Friedrichs method and uses reconstruction to obtain higher order accuracy. A third type of methods are the so called ENO (essentially non-oscillatory) methods, where the stencil is chosen adaptively

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to obtain a solution with as little oscillations as possible. Even if the formal order of accuracy of the high resolution scheme may not be higher than that of the corresponding basic schemes, the performance is in general much better. However, there are still many cases where the performance is not satisfactory. The high resolution methods are complex and difficult to analyze. Hence, further improvements would be enhanced by the development of better methods of analysis.

One example of a numerical artifact is the post shock oscillations which are generated by many shock capturing schemes when computing numerical solutions containing slowly moving shocks, see e.g., [1,17,33], and references therein. Another problem is that high order shock capturing schemes often yield numerical solutions which are only first order accurate downstream of shocks, see e.g., [7–9,11,10,18,31,30]. For reasonable mesh sizes, the first order term is small in many cases. However, in application where the small scale behavior is of significance, such as aeroacoustics, the degeneration to first order accuracy can be troublesome. The first order error term can be explained in the following heuristic way. In the shock layer, the point wise errors are $\mathcal{O}(1)$. The information that passes through the shock layer interacts with the shock, and the interaction time is $\mathcal{O}(\Delta t)$. The result is a first order downstream error even in high resolution schemes. This phenomena has been analyzed in detail in [10,18,31,30]. The analysis was based on the assumption that the numerical solution can be modeled by a continuous function which is the solution of a PDE, a so called modified equation, see [19]. The modified equation was analyzed using matched asymptotic analysis. The analysis shows that the downstream error term is related to the shape of the numerical solution in the shock layer. In other words, the size of the downstream error is related to the path in phase space connecting the left and right shock states. For time-dependent solutions of systems of hyperbolic conservation laws in one space dimension, it was shown in [31] that e.g., if the numerical solution follows a straight line in phase space, then the first order error term is eliminated. By designing a special matrix valued viscosity, we could verify in computations that the first order error term was eliminated.

The technique of modified equations is a common tool widely used for the analysis and development of numerical methods, see [13,34], and references therein. The technique is, e.g., often used to analyze dispersive and dissipative errors in linear finite difference schemes for hyperbolic conservation laws. The analysis mentioned above is one example that shows the usefulness of a continuous model for the discrete, numerical solution. Often it is much easier to analyze a PDE than the discrete scheme itself. However, there is surprisingly little literature concerning the validity and the scope of the method of modified equations [13,34]. The derivation of the modified equation is based on the assumption that the numerical solution can be modeled by a continuous equation. The modified equation is then derived by Taylor series expansions. To do this is straightforward for simple schemes as e.g., the Lax–Friedrichs scheme. Concerning the use of modified equations as model for numerical solutions in the vicinity of shocks, there are still many open questions. The first and most important question is if/when it is appropriate to model the discrete numerical solution by a continuous function. In other words, when does a stable so called continuum shock profile exist? This is not known for shock capturing methods in general, especially not for high resolution methods. Below, we will make a precise definition of the concept continuum shock profile, and give a survey of the research in this area. Secondly, it is difficult to derive a modified equation for a high resolution scheme, since these methods in general are complicated and involve non-linear limiters. In the analysis mentioned above, we used a modified equation which corresponds to a generic second order shock capturing scheme. To be able to compare different shock capturing methods, one would need an exact modified equation for the specific schemes. A related problem is that due to the strong gradients in the shock layer, it is not obvious, even for relatively simple scheme, which terms in the Taylor expansions that will be dominant, see e.g., [17].

Before we proceed, let us introduce some notation. We consider a system of hyperbolic conservation laws

$$q_t + f(q)_x = 0, \quad t \geq 0, \quad x \in \mathbf{R}, \quad q = q(x, t) \in \mathbf{R}^n. \quad (1)$$

We consider Riemann initial data admitting a single k -shock traveling with speed s , i.e.

$$q(x, 0) = \begin{cases} q_L, & x < 0, \\ q_R, & x > 0, \end{cases} \quad (2)$$

where q_L and q_R are chosen such that the Rankine–Hugoniot condition

$$s(q_R - q_L) = f(q_R) - f(q_L)$$

and the Lax entropy condition

$$\begin{aligned} \lambda_k(q_R) &> s > \lambda_k(q_L), \\ \lambda_j(q_L) &< s \quad \text{and} \quad \lambda_j(q_R) < s \quad \text{for } j = 1, 2, \dots, k-1, \\ \lambda_j(q_L) &> s \quad \text{and} \quad \lambda_j(q_R) > s \quad \text{for } j = k+1, k+2, \dots, n, \end{aligned}$$

are satisfied. Here, the eigenvalues of $f'(q)$ are denoted by $\lambda_j(q)$, $j = 1, 2, \dots, n$ and ordered in increasing order. Hence, the solution of (1) is

$$q(x, t) = \begin{cases} q_L, & x < st, \\ q_R, & x > st. \end{cases}$$

Eq. (1) with initial data (2) is discretized by some conservative numerical scheme

$$v_j^{n+1} = \mathcal{L}[v^n]_j, \quad (3)$$

where $v_j^n \approx q(j\Delta x, n\Delta t)$, $n = 0, 1, \dots$, $j = 0, \pm 1, \pm 2, \dots$.

Let us now introduce the concept continuum shock profile for a numerical method. A traveling wave $\phi(x - st)$ is a continuum shock profile associated to scheme (3) if

$$v_j^n = \phi(j\Delta x - sn\Delta t + \xi\Delta x)$$

for $n = 0, 1, 2, \dots$, $j = 0, \pm 1, \pm 2, \dots$ and any given $\xi \in [0, 1]$. Note that if $\eta = s\Delta t/\Delta x$ is rational, the discrete solution will repeat itself after a finite number of steps.

The existence and stability of continuum shock profiles for general numerical schemes is difficult to prove, and is still an open question in most cases. Many authors have addressed the issue, but theoretical results are limited. For rational η , the existence of continuum shock profiles has been proven for shocks of any strength for scalar equations for monotone schemes in [16], the Lax–Wendroff scheme in [32], and a class of monotonicity preserving schemes in [12]. For systems of equations with weak shocks with rational η the existence of continuum shock profiles has been proven for diffusive first-order schemes satisfying a non-resonance condition, e.g., Godunov's scheme and the Lax–Friedrich scheme, in [24]. The result in [24] was extended to certain third order schemes in [25]. Liu and Yu [22,23] consider systems with shocks where η is irrational but satisfies the Diophantine property. Also in this case, one has a recurring property. The shock must be weak when some constants connected to the Diophantine property are small. The existence and stability of shock profiles are proven for the same class of schemes as in [24]. As a mean to avoid the limitations of rational speed or Diophantine speed, the semi-discrete problem is studied by Benzoni-Gavage, Hout and Rousset [2–5]. Existence and stability of shock profiles is obtained for the semi-discrete versions of the schemes treated in [24] for weak shocks. Bultelle et al. [6, p. 2285] discuss the possibility that the sequence of discrete solutions does not converge to a continuum shock profile, though it remains bounded. They suggest the possibility that the sequence of discrete solutions tends towards a stable limit cycle, but do not give any specific example, or any other evidence, that such situations actually do occur. An overview of the theoretical results concerning existence and stability of discrete shock profiles is given in [28].

Numerical studies of the existence of continuum shock profiles for numerical methods are rare in the literature. A convenient tool for numerically investigating the behavior of a numerical scheme in the shock layer is to track the trajectories of the mesh points in phase space as the shock traverses the mesh. By superposing the tracks of all mesh points one obtains a clear picture of the behavior of the numerical solution in the shock region. Since the phase space orbits obtained are complex, this method is best suited for system of equations with two equations. In the numerical studies mentioned below, the numerical experiments are done using either the isothermal Euler equations, the p -system or the shallow water equations. Also, all the numerical investigations mentioned below focus on investigating oscillations around slowly moving shocks. Numerical phase space orbits were first used by Roberts [27] who studied Godunov's scheme, the Roe scheme and the Osher scheme. The phase space plots he obtained show that the point values in the numerical solution fall on a well defined single curve in phase space. Also the experience from later studies is that for the schemes studied the numerical solution follows a single well defined curve in the phase space. Arora and Roe [1] studied some first and second order schemes, such as the Roe method and Lax–Wendroff method, as well as two flux-limited second order schemes. They found that the phase space orbits for the flux-limited schemes were similar to the phase space orbit for the underlying first order schemes. In [1] the phase space plots are presented using

the characteristic variables as coordinates. Karni and Čanić [17] investigated the first order upwind scheme and the Lax–Friedrichs scheme. They compare the phase space plot of the numerical solutions with corresponding plots of the modified equations. For a model problem, they show how oscillations could be eliminated by introducing an entropy fix, which is equivalent to increasing the amount of numerical viscosity. Stiriba and Donat [33] present phase space plots for a slow shock for some basic schemes, including the first order Marquina scheme. Also here, an entropy fix which increases the amount of viscosity is suggested in order to reduce the oscillations. The phase space orbits presented in [17,27,1] all seem to be smooth. In [33] however, the orbits presented for the van Leer scheme and especially the orbit for the Marquina scheme are non-smooth with distinct jerks.

In this paper we have numerically studied discrete shock profiles for a range of high resolution schemes. The main result is that we have found examples where no continuum shock profile exists, but the point values in the shock transition appear to converge towards a stable limit cycle, as anticipated in [6]. The paper is organized as follows. In Section 2 we report on our numerical experiments, and in Section 3 we summarize our conclusions.

2. Numerical experiments

We have experimentally investigated the numerical shock profiles for a class of second order Riemann solver based flux-limiter methods, a class of high resolution central schemes and a second and a third order ENO method. We numerically solved Riemann problems with solutions consisting of a single shock. Phase space orbits for the numerical shock profiles were obtained by, in each time step, plotting the point values of the solution in phase space. The phase space plots shown below were generated from discrete solution at hundreds of different points in time. In each time step, there might be as few as two to four data points in the shock profile. Hence, the points from one time step is not enough to resolve the numerical orbit, but when the solutions from many time steps are collected, a clear picture of the numerical orbit appears. To avoid initial transients we omitted the solution from the first couple of hundred time steps. The number of time steps it takes for the shock profile to settle varies with e.g., the method and the CFL number. We have used both conserved variables and characteristic variable as coordinates. In the experiments, we mainly focused on the isothermal Euler equation, but also made some experiments with the p -system and the shallow water equations.

Before we proceed, let us introduce the test cases for which we will present numerical experiments. We consider the isothermal Euler equations,

$$q_t + f(q)_x = 0,$$

where

$$q = \begin{pmatrix} \rho \\ m \end{pmatrix}, \quad f(q) = \begin{pmatrix} m \\ m^2/\rho + \rho a^2 \end{pmatrix}, \quad a = \text{const.}$$

The constant a is taken to be 1. The density ρ and the momentum m are related by $m = \rho u$, where u is the velocity. The wave speeds (i.e. the eigenvalues of the Jacobian $f'(q)$) are

$$\lambda_{1,2} = u \mp a,$$

and the characteristic variables are

$$C^\mp = \ln \rho \mp \frac{u}{a}.$$

In Table 1 we define four Riemann problems for the isothermal Euler equations. These four Riemann problems are used in the numerical experiments presented below, and referred to as test case T1–T4. Note that since we only consider Riemann problems where the solution is a single shock, the problems are fully defined by the three variables ρ_L , m_L and ρ_R . The slowly moving shock T1 is used in computations also in [33], where phase space plots are shown for other methods than we have used.

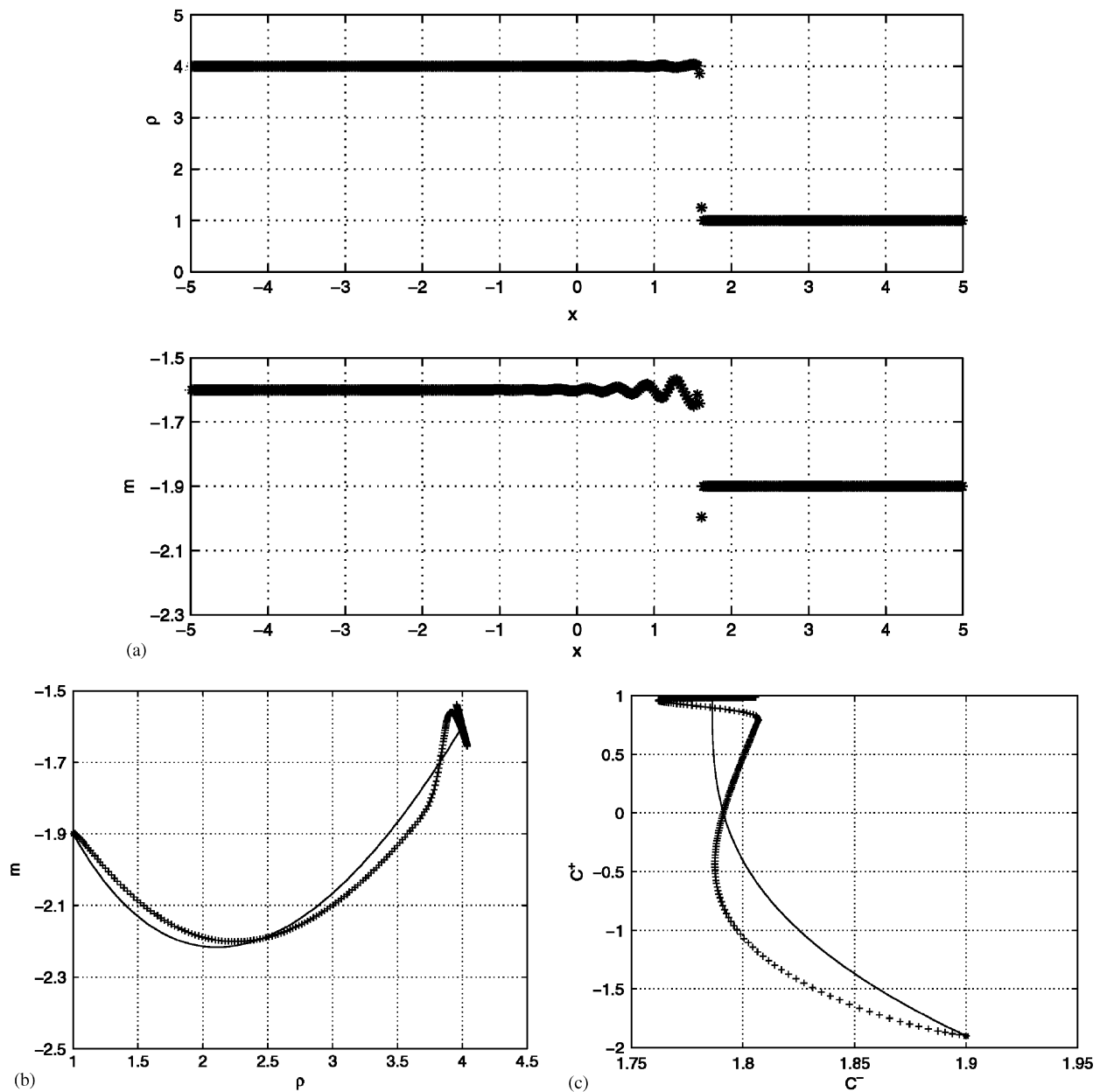
2.1. Clawpack wave propagation schemes

In the software package Clawpack by LeVeque and co-workers [20], a class of second order flux limiter methods are implemented. We have used Clawpack, version 4.2. The code and documentation is freely available [20], and

Table 1

Data defining the test cases for which we will present numerical experiments

	ρ_L	m_L	ρ_R	m_R	s	
T1	4	-1.6	1	-1.9	0.1	2-shock
T2	4	0	1	-1.5	0.5	2-shock
T3	1	1	6	≈ -6.25	≈ -1.45	1-shock
T4	40	14	44	≈ 11.20	≈ -0.70	1-shock

Note that the Riemann problems are fully defined by the three variables ρ_L , m_L and ρ_R .Fig. 1. Solution of test case T1 obtained by Godunov's method, $\Delta x = 0.025$ and $\Delta t / \Delta x = 0.32$: (a) solution in physical space; (b) phase space orbit, ρ - m plane; (c) phase space orbit, C^- - C^+ plane.

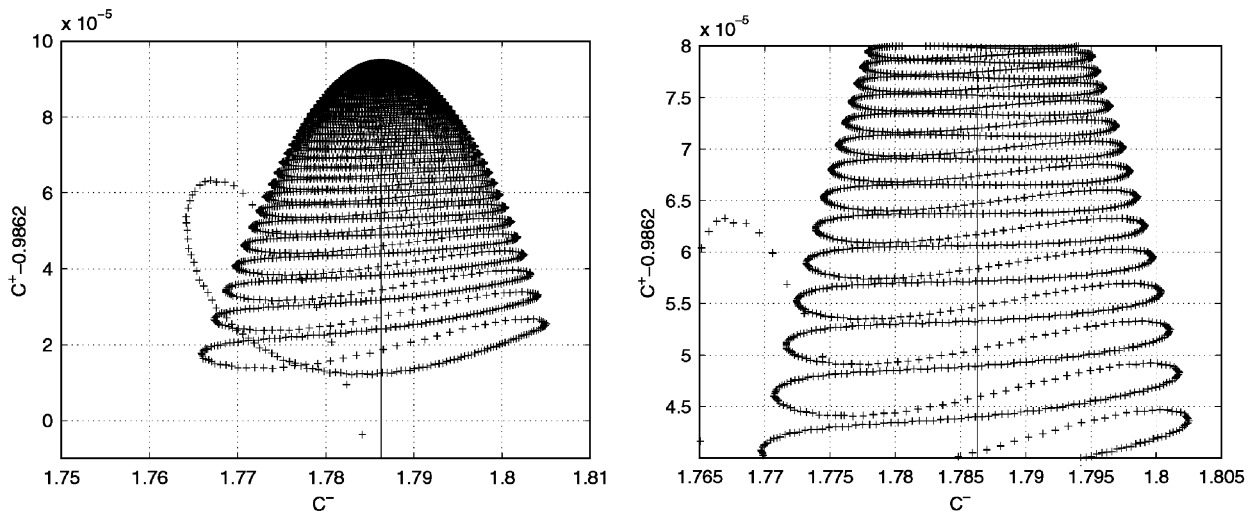


Fig. 2. Zooms of Fig. 1(c).

an extensive description of the numerical methods in Clawpack is given in [19]. The methods are based on the first order Godunov's method. By adding correction terms, more accurate methods are obtained. The most basic form of correction term gives the second order Lax–Wendroff method. To obtain high resolution methods, limited versions of the Lax–Wendroff correction terms are used. We have investigated how the choice of limiter affects the phase space orbits of the numerical shock profile. We have used the basic Godunov's scheme, the Lax–Wendroff scheme, and high resolution schemes obtained by using the Minmod-, Superbe-, van Leer-, and MC- (monotonized centered) limiter. We will denote the high resolution methods CLAW-MM, CLAW-SB, CLAW-vL, and CLAW-MC, respectively. In Clawpack, the user can choose different limiters for each characteristic field. In all cases, we have used the same limiter in all characteristic fields. We have used Roe's approximate Riemann solver for the isothermal Euler equations which is provided in Clawpack. In all computations, we have used constant space and time steps. Before we summarize our conclusions from numerical experiments, let us study the results from the test cases T1, T4, and T3 in some detail.

Let us first consider the slowly moving shock in test case T1. Note that test case T1 is an example of a slowly moving shock where post shock oscillations are generated by many shock capturing schemes, and that T1 is used as a test case also in [33]. Results from numerical experiments with Godunov's method, the Lax–Wendroff method, and the CLAW-MC method are presented in Figs. 1, 2, 3, and 4, respectively. In the computation, $\Delta x = 0.025$ and $\Delta t / \Delta x = 0.32$ were used. The solution obtained by Godunov's method has oscillations on the downstream side of the shock wave. The oscillations decay rather rapidly away from the shock. In the momentum component, the oscillations are large, and much smaller in the density component. The phase space orbit for the solution is very clear. All points in the numerical solutions seem to fall on a single, smooth curve in phase space with high accuracy. Also in the phase space orbit, the oscillations downstream of the shock are seen. The solution oscillates back and forth around q_L . In phase space, the oscillatory part of the solution is very compressed, and one must zoom very close to distinguish the separate oscillations. However, zooming in on the phase space orbit for the characteristic variables around the oscillations shows how extremely well defined the curve is, see Fig. 2. The picture is qualitatively the same also when conserved variables are used as coordinates. Note that on the upstream side of the shock, there is a momentum spike. In physical space the momentum spike, which is not present in the mathematical solution of the Riemann problem, appears rather strange. However, the plot of the phase space orbit of the numerical solution helps us to understand this behavior. The numerical orbit in phase space is non-monotone in the momentum component, see Fig. 1(b). This also applies for the corresponding Hugoniot curve. The numerical phase space orbit and the Hugoniot curve are relatively close. Let q_R denote the upstream state in the Riemann problem T1. Hence, Riemann problems where the solution jumps from q_R to $\tilde{q}_L$, where $\tilde{q}_L$ is a state on the Hugoniot locus for T1 sufficiently close to q_R , will have a decrease in the momentum. Numerical experience, from the present study and the numerical studies mentioned above, shows that the closeness between the numerical phase space orbit and the Hugoniot locus is general. Hence, a momentum spike will appear

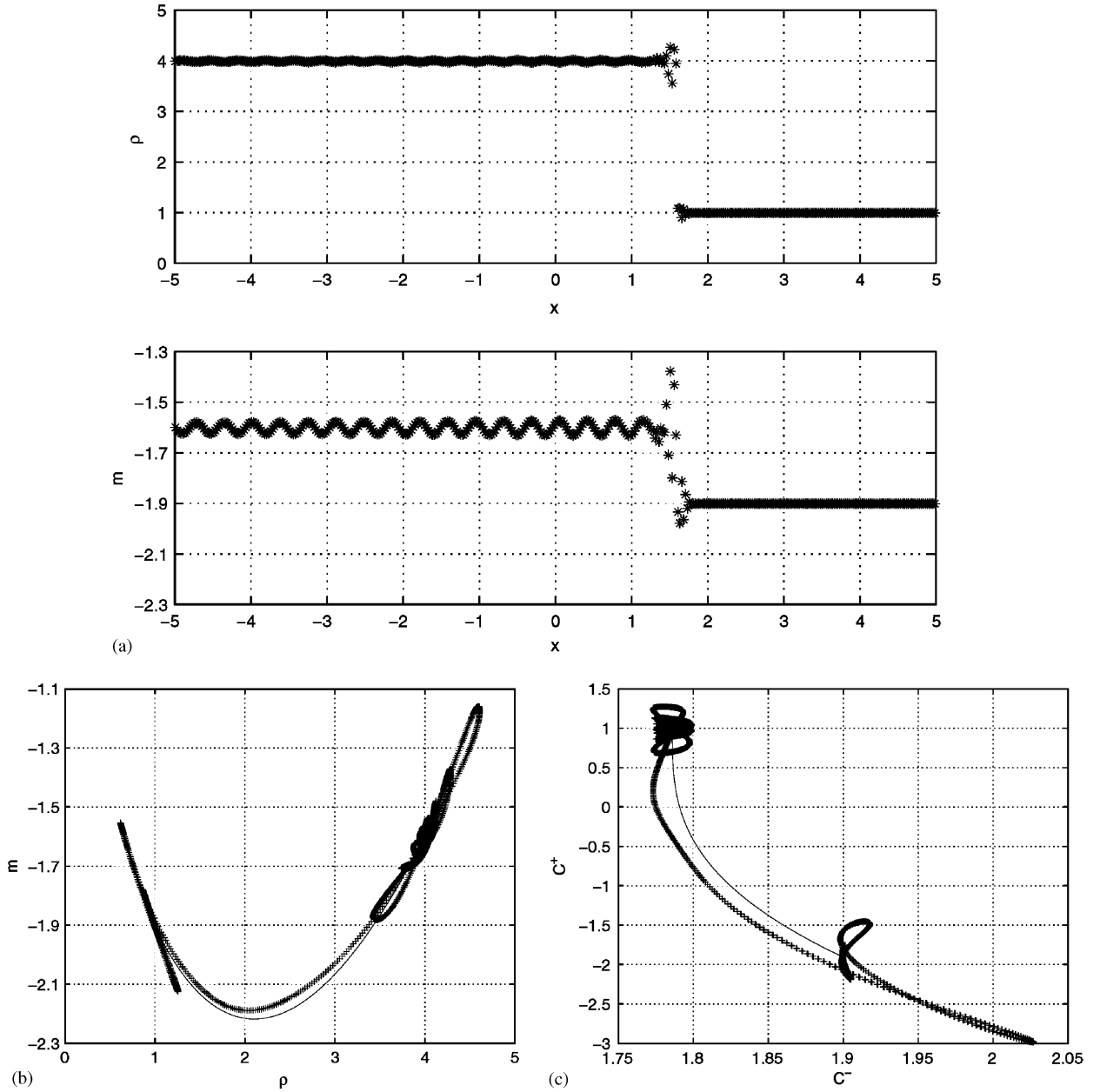


Fig. 3. Solution of test case T1 obtained by the Lax–Wendroff method, $\Delta x = 0.025$ and $\Delta t/\Delta x = 0.32$: (a) solution in physical space; (b) phase space orbit, ρ – m plane; (c) phase space orbit, C^- – C^+ plane.

in the solution when the Hugoniot locus is non-monotone in the momentum component. The “unsteadiness” of the momentum spike, see [33,21], is due to that at each time, there are only a few points in the numerical profile. Hence, the amplitude of the spike appears to fluctuate, while the numerical phase space orbits reveal that the momentum spike is part of a very steady structure.

The Lax–Wendroff solution has over shoots on both sides of the shock. On the downstream side, they persist as a long wavy tail. The phase space orbit is also wild, but follows a well defined smooth curve in phase space.

The CLAW-MC solution behaves as Godunov’s method in the shock region, but downstream of the shock the solution shows slowly decaying oscillations. Again, the phase space orbit is very distinct. When plotted in conserved variables,

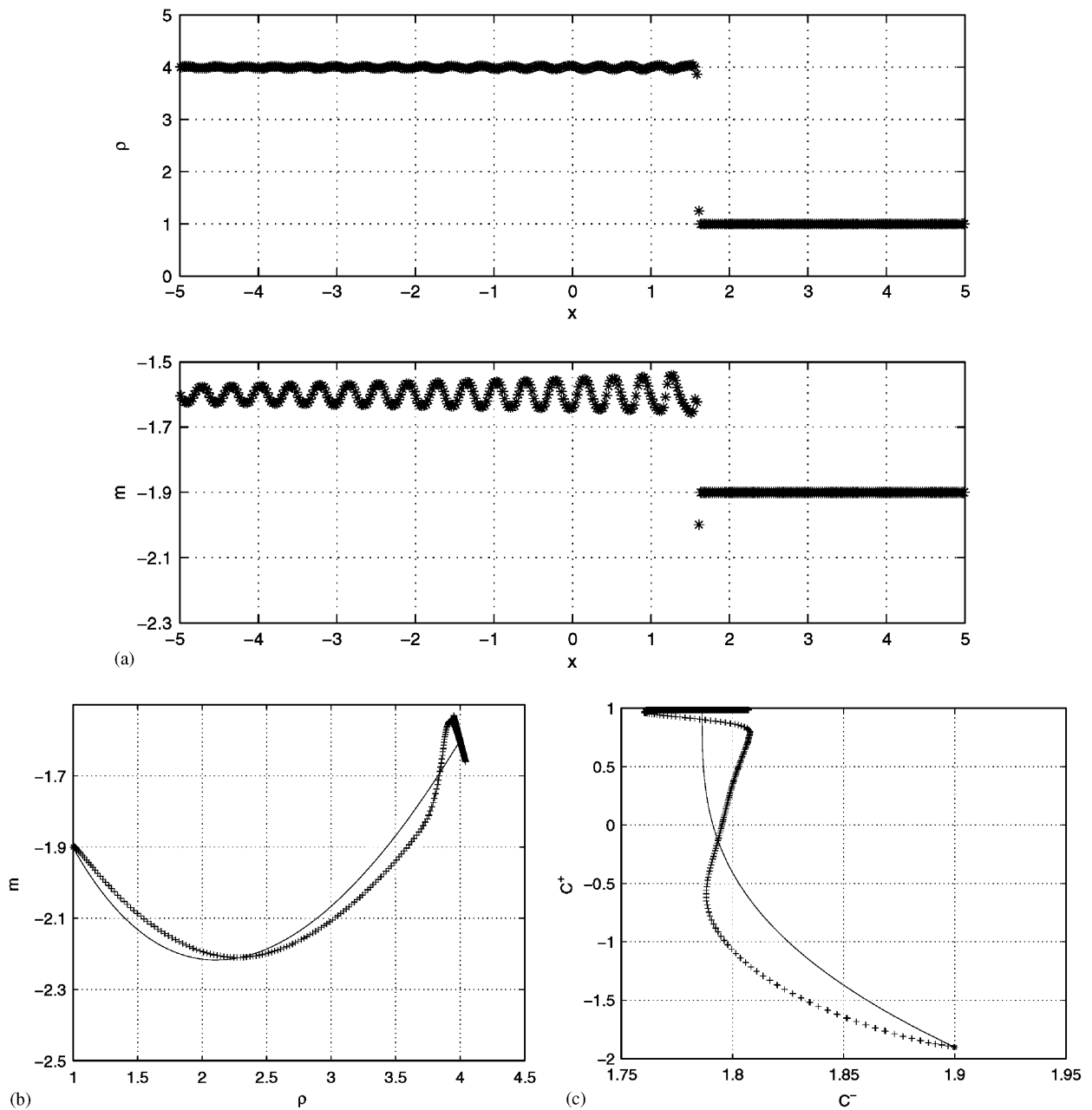


Fig. 4. Solution of test case T1 obtained by the CLAW-MC method, $\Delta x = 0.025$ and $\Delta t/\Delta x = 0.32$: (a) solution in physical space; (b) phase space orbit, ρ - m plane; (c) phase space orbit, C^- - C^+ plane.

the orbit is very similar to the phase space orbit for Godunov's method. However, when plotted in characteristic variables, one can see that the CLAW-MC orbit is less smooth. The phase space orbits are, however, very similar. Computations for this test case with the other high resolution schemes in Clawpack give results which are very similar to the result for the CLAW-MC.

This computation shows that one cannot see from the phase space orbit if there will be downstream oscillation after slowly moving shocks. In the above example, two methods have phase space orbits which are very close, but totally different behavior concerning downstream oscillations. For this test case, the amount of oscillations that are generated

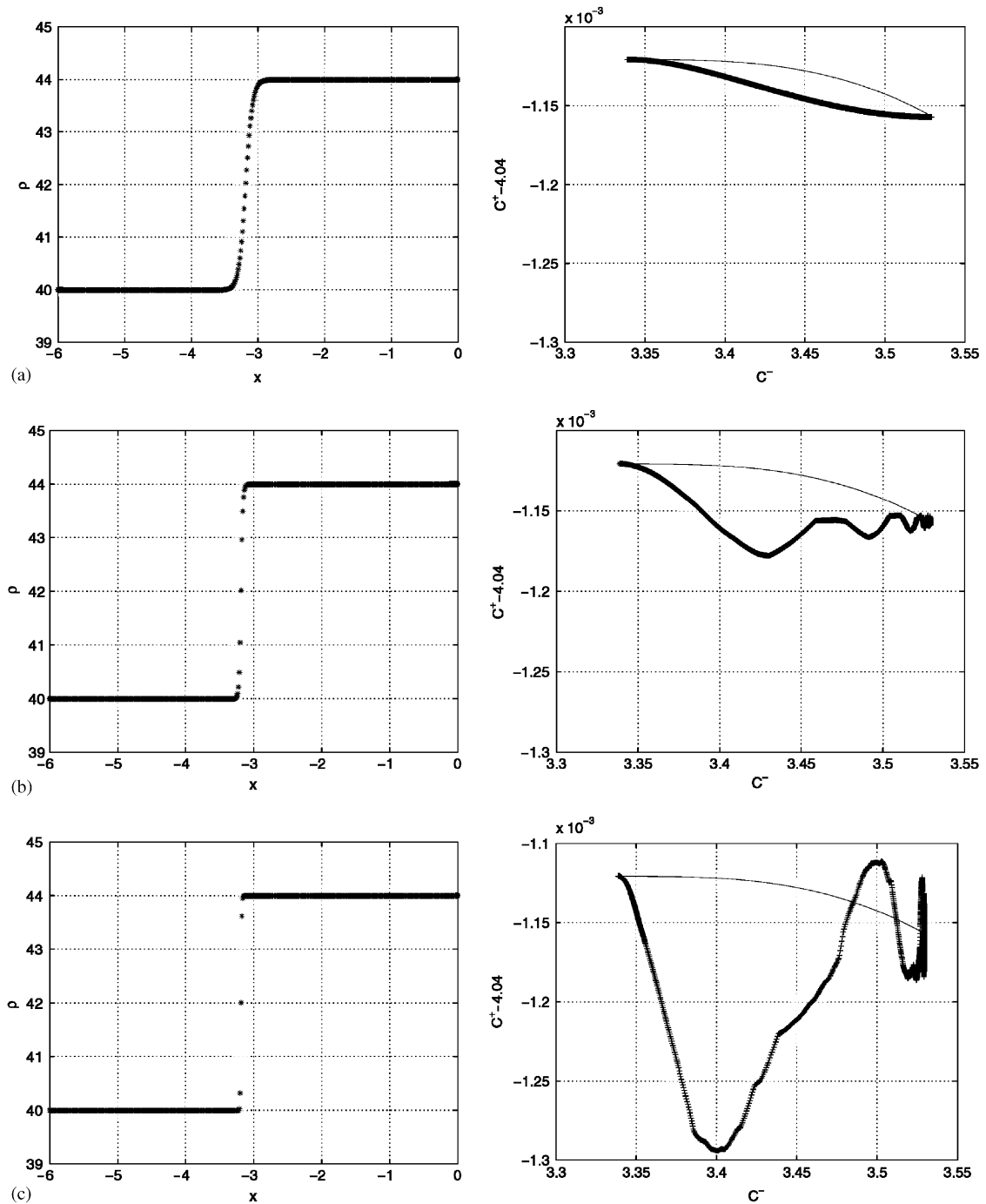


Fig. 5. Results for test case T4 for three different methods. To the left, the density as a function of x , to the right, phase space orbits plotted in the $C^- - C^+$ plane. In all computations, $\Delta x = 0.0125$ and $\Delta t/\Delta x = 0.64$: (a) Godunov's method; (b) CLAW-MM; (c) CLAW-SB.

is similar for both the methods. The different behavior occurs because of different damping properties in the methods. The fact that the phase space orbit for the limited second order schemes are very similar to Godunov's scheme is not surprising, since in the vicinity of the shock, we expect the high resolution scheme to reduce to the underlying first order scheme. Outside the shock region, however, the high-resolution schemes are second order accurate, with little damping, while Godunov's method is first order accurate with strong damping of oscillations.

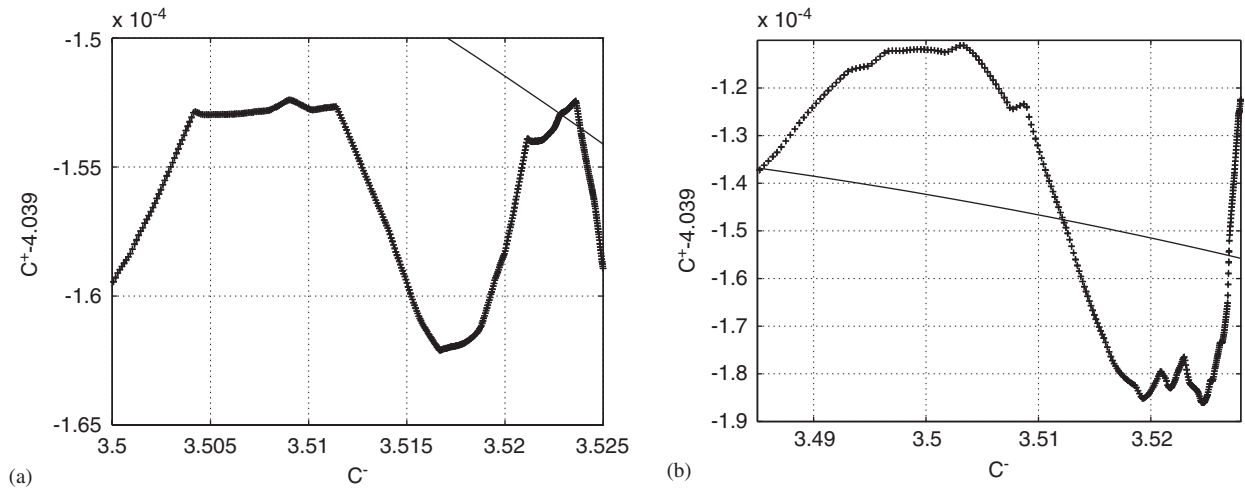


Fig. 6. Close-up of the phase space orbits in Fig. 5(b,c). Note the different scale in the two pictures: (a) CLAW-MM; (b) CLAW-SB.

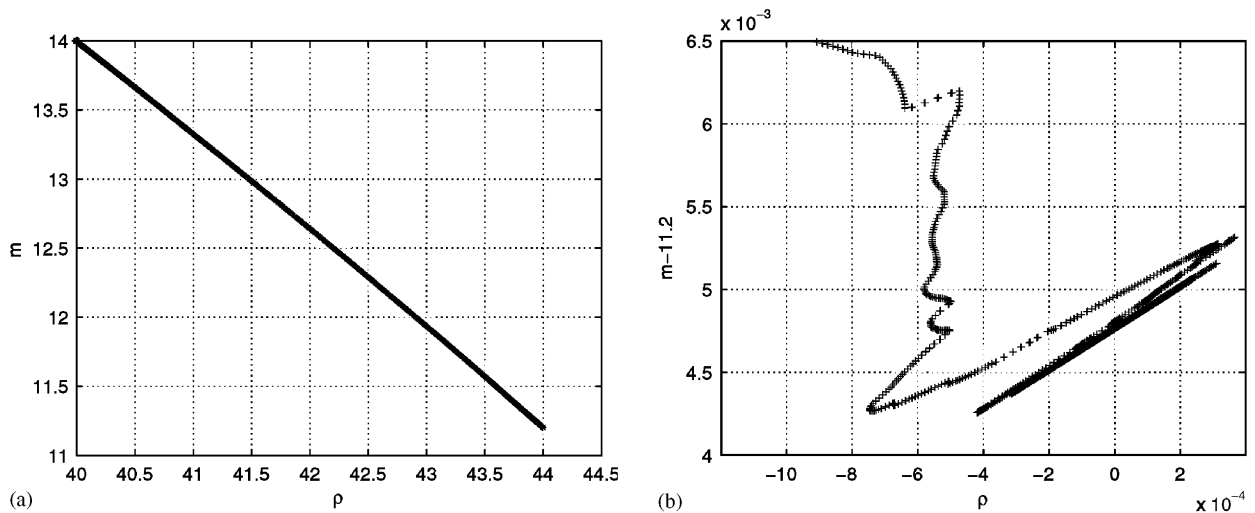


Fig. 7. The phase space orbit plotted in the ρ - m plane for test case T4 using the CLAW-SB method with $\Delta x = 0.0125$ and $\Delta t/\Delta x = 0.64$: (a) overview; (b) zoom.

Let us proceed to test case T4, which is a relatively weak shock. Here, the difference between the different limiters are seen clear in the numerical solution when plotted in physical space. As expected, the Superbe limiter gives the sharpest shock wave, followed by the MC-, the van Leer- and the Minmod limiter. The pure Godunov's scheme gives a very smeared numerical solution. In Fig. 5, the ρ -component of the solution is plotted, as well as the phase space orbit in characteristic variables for Godunov's method, the CLAW-MM method and the CLAW-SB method. When solving T4, we used $\Delta x = 0.0125$ and $\Delta t/\Delta x = 0.64$. For all methods, the phase space orbits follow a single, well defined curve. Also in phase space, the orbits are ordered in "sharpness order". Hence, the orbit for the CLAW-MM method is closest to the orbit for Godunov's method, followed by CLAW-vL, CLAW-MC and CLAW-SB, in this order. The phase space orbit for Godunov's method is smooth, while the orbits for the limited methods have jerks and corners and a lot of fine-scale details, see Fig. 6. When plotted using conserved variables, the phase space orbits at a first glance seem to be totally similar for the different methods. One must zoom in close to the orbit to see the fine-scale features, see Fig. 7.

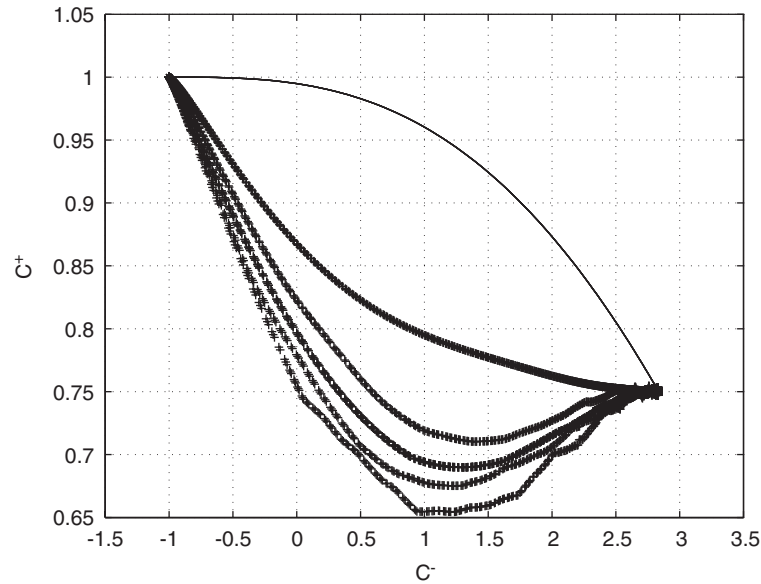


Fig. 8. Phase space orbits for test case T3 using several different methods. From below: CLAW-SB, CLAW-MC, CLAW-vL, CLAW-MM, and Godunov's method. In all computations, $\Delta x = 0.025$ and $\Delta t/\Delta x = 0.2$. The phase space orbits are plotted in characteristic variables.

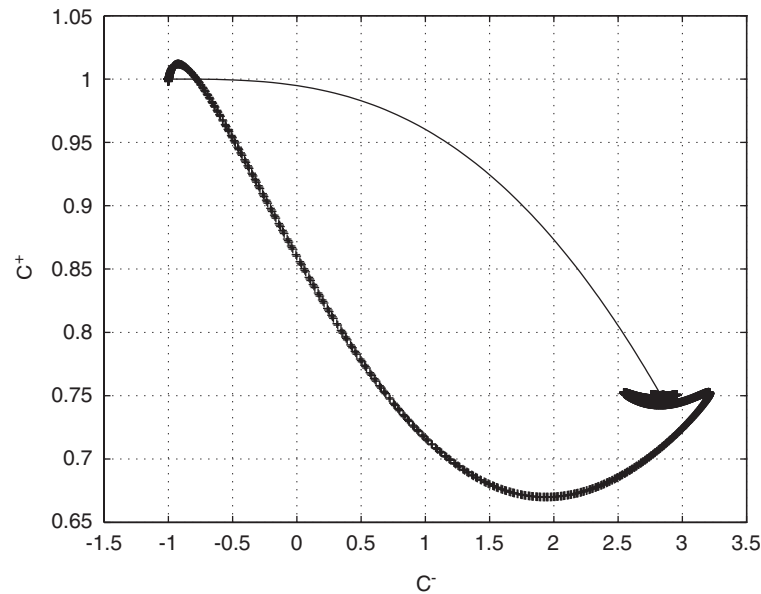


Fig. 9. Phase space orbit for test case T3 using the Lax-Wendroff method with $\Delta x = 0.025$ and $\Delta t/\Delta x = 0.2$. The phase space orbit is plotted in characteristic variables.

Finally, let us consider test case T3, which is a strong and relatively fast shock. In the computations, $\Delta x = 0.025$ and $\Delta t/\Delta x = 0.2$. The Lax-Wendroff method gives a solution with overshoots close to the shock. The oscillations decay rapidly. In physical space, the solutions of high resolution schemes have sharp profiles and hardly noticeable oscillations. The solution obtained by Godunov's method is more smeared. In Figs. 8 and 9, phase space orbit using characteristic variables as coordinates are shown. Again, the same ordering as for T4 is seen. Zooming in on the phase space plots for the CLAW-vL solution of test case T3, one sees that close to the downstream side, the phase space orbits seem to be only piecewise continuous, see Fig. 10. As mentioned in the Introduction, the point values that the numerical

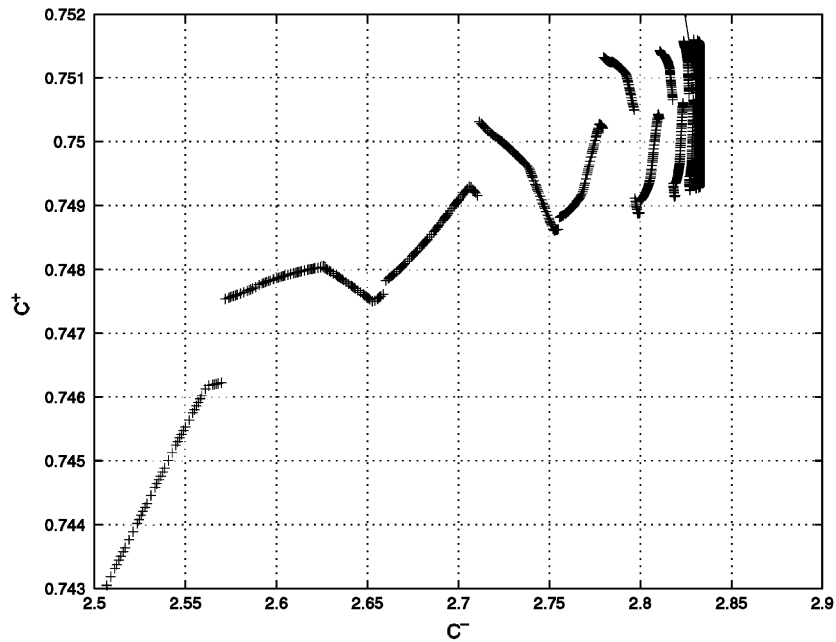


Fig. 10. Zoom of the phase space plot in Fig. 8 for the CLAW-vL solution of test case T3, with $\Delta x = 0.025$ and $\Delta t/\Delta x = 0.2$.

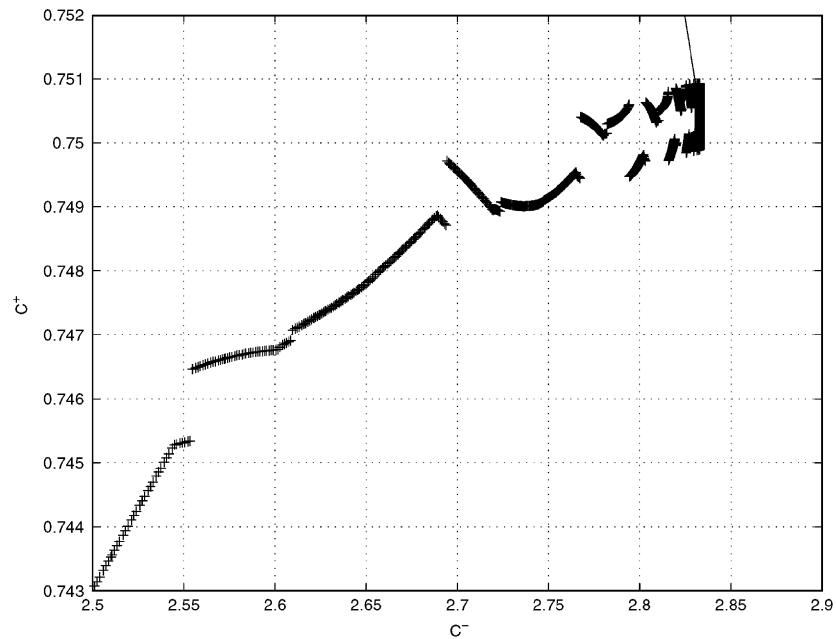


Fig. 11. Zoom of the phase space plot for the CLAW-vL solution of test case T3, with $\Delta x = 0.025$ and $\Delta t/\Delta x \approx 0.188$.

solution will obtain depend on $s\Delta t/\Delta x$. In order to investigate if the jumps in the orbit are just a discrete effect that arises only for certain $\Delta t/\Delta x$, we numerically investigated how the orbit changes as $\Delta t/\Delta x$ changes. The conclusion is that the piecewise discontinuous behavior remains. Also the orbit itself changes, sometimes relatively much for a small change in $\Delta t/\Delta x$. One example is shown in Fig. 11, where $\Delta t/\Delta x \approx 0.188$. The orbit clearly differs from the orbit in Fig. 10, where $\Delta t/\Delta x = 0.2$, while the piecewise discontinuous behavior remains. Hence, we conclude that although

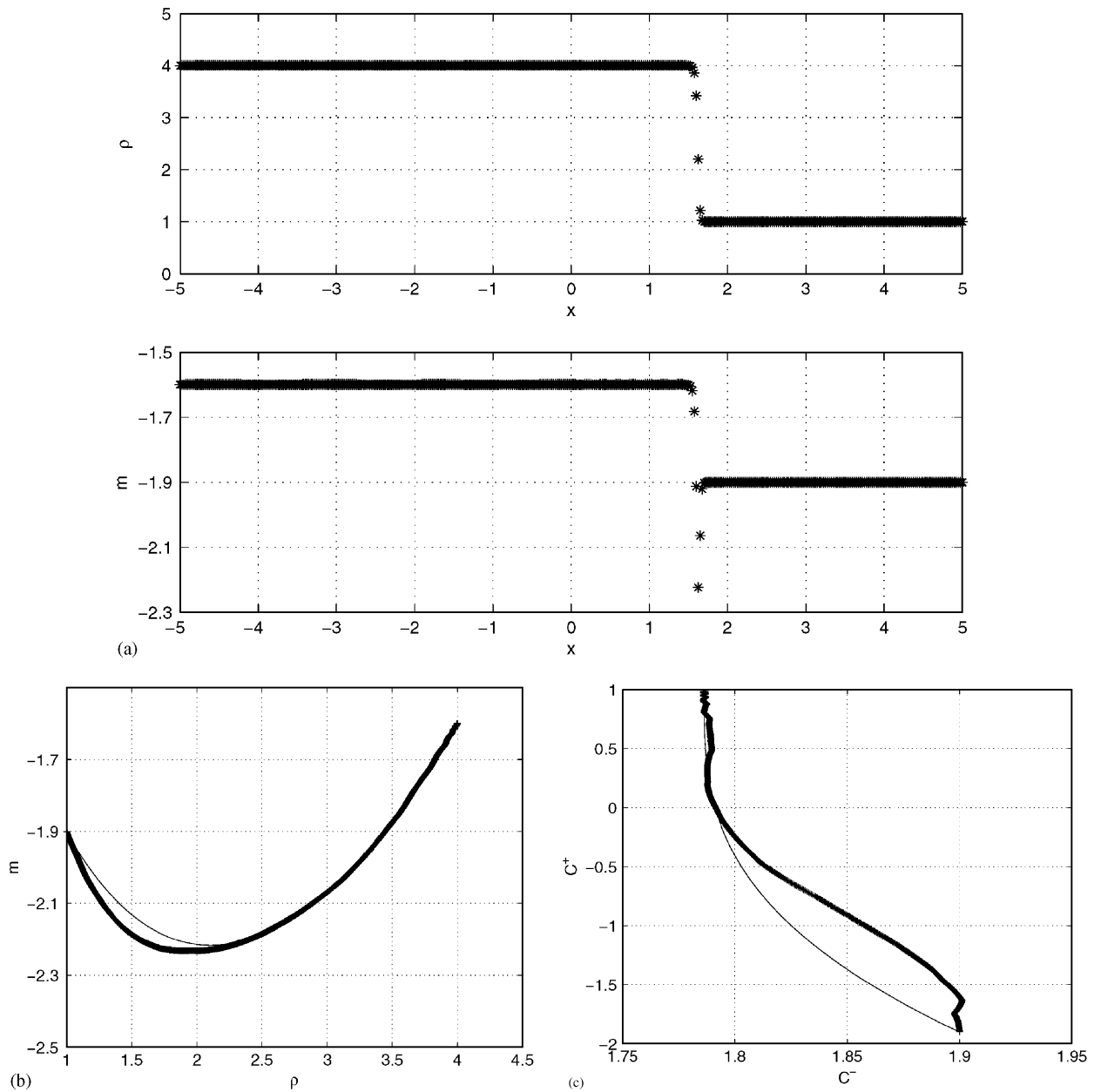


Fig. 12. Solution of test case T1 obtained by STG with $\Delta x = 0.0025$ and $\Delta t/\Delta x = 0.32$: (a) solution in physical space; (b) phase space orbit, ρ - m plane; (c) phase space orbit, C^- - C^+ plane.

the solution of test case T3 appears to be only piecewise continuous, the numerical experiments indicate the existence of a continuum shock profile for the CLAW-vL method.

Let us summarize the conclusions from the experiments using the numerical methods in Clawpack. Our numerical experiments indicate the existence of a continuum shock profile for all the investigated methods. After an initial transition period, the numerical phase space orbits obtained are always very distinct and remain faithfully. Generally, the data points from different time steps follow a common path with at least six digit accuracy. The only situation when it is not clear that the point follows a common path is very, very close to the end state where oscillations occur. There the picture is blurry, and it is not clear that all points lie on a common path. One reason that the “blurriness” is more

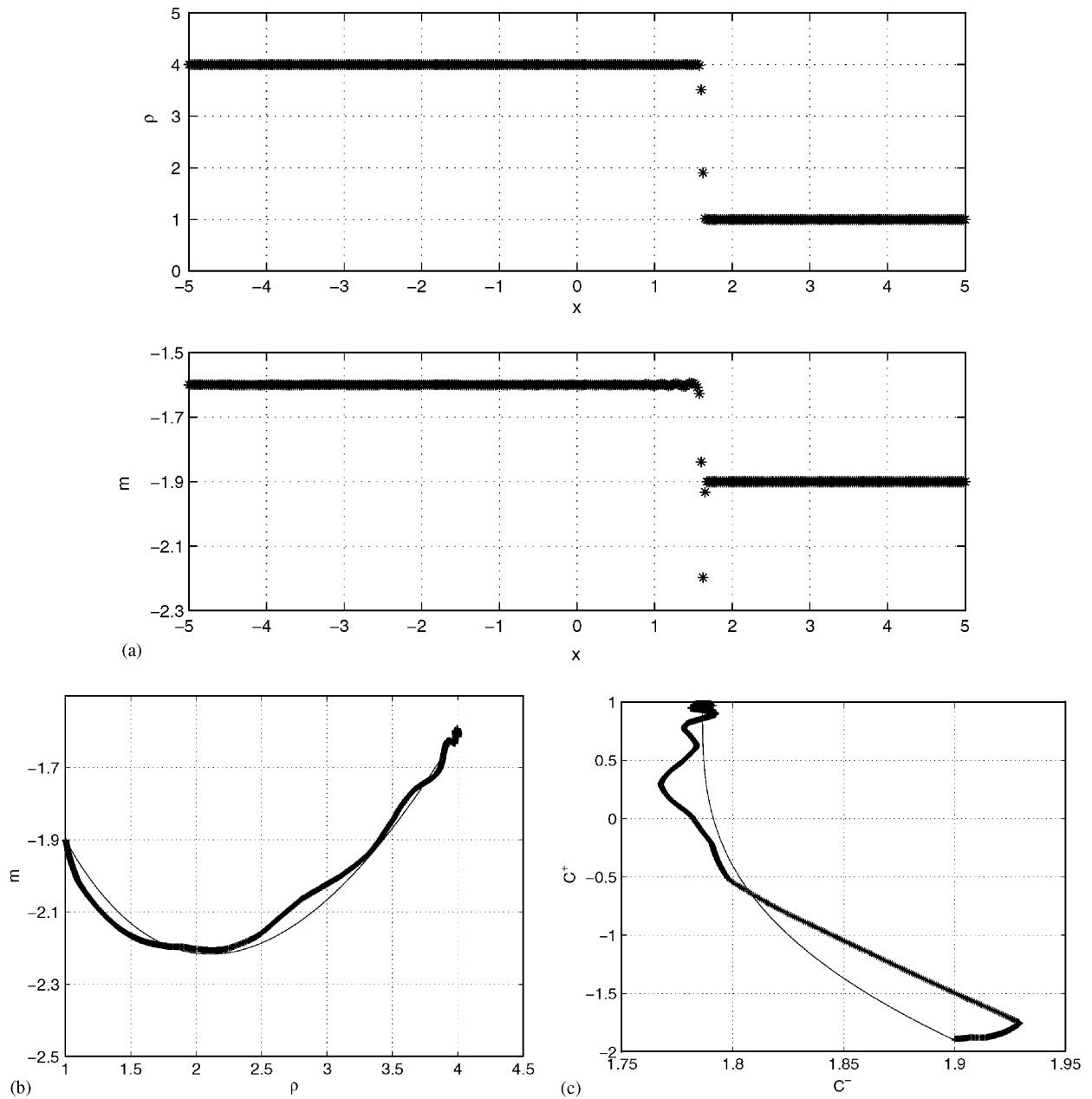


Fig. 13. Solution of test case T1 obtained by STG2 with $\Delta x = 0.0025$ and $\Delta t/\Delta x = 0.32$: (a) solution in physical space; (b) phase space orbit, ρ - m plane; (c) phase space orbit, C^- - C^+ plane.

clearly seen in the oscillations close to an end state is probably that there are so many more data points there. It is much more difficult to see “blurriness” in the middle of the numerical orbit, where there are relatively few data points.

Generally, the numerical profiles follow the Hugoniot curve rather closely when plotted in conserved variables. In conserved variables, the difference is small, often hardly noticeable, between the different methods. When plotted in characteristic variables, the difference is seen much clearer. The character of the phase space orbit is highly dependent on the specific flow case. For Godunov’s method and the Lax–Wendroff method (where no limiting is involved) the curve seems to be always smooth. Also for the high resolution schemes, the phase space orbit is sometimes smooth. However, often the orbits for the limited methods have jerks and corners and a lot of fine-scale details.

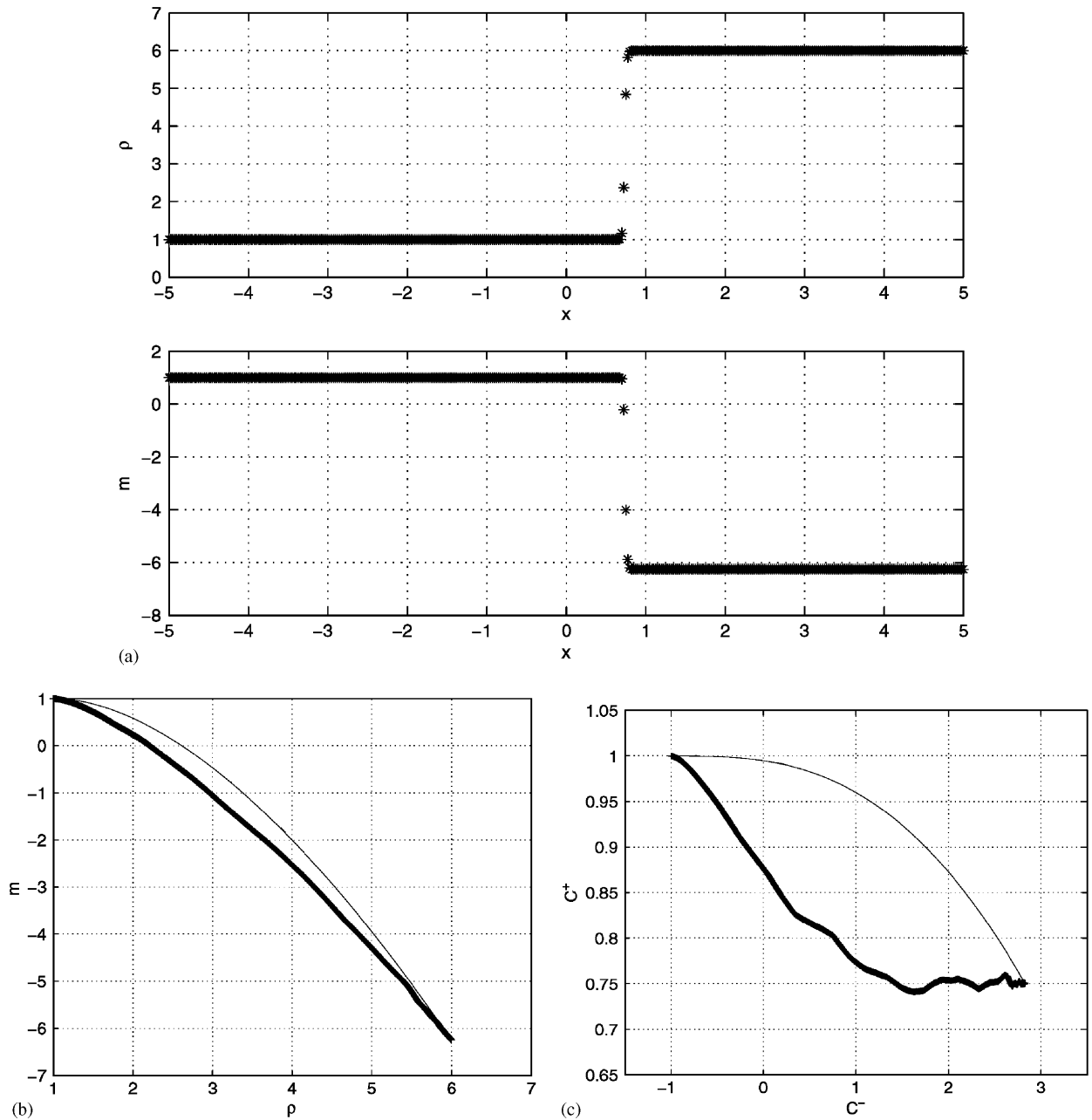


Fig. 14. Solution of test case T3 obtained by STGU with $\Delta x = 0.0025$ and $\Delta t / \Delta x = 0.2$: (a) solution in physical space; (b) phase space orbit ρ - m plane; (c) phase space orbit, C^- - C^+ plane.

The phase space orbit depends on the CFL-number. How strong this dependence is varies a lot from case to case. Some solutions are very sensitive to changes in CFL-number, while in other cases the sensitivity is very low.

It seems it is in general reasonable to model the numerical solution of the Clawpack methods by a piecewise smooth continuous function. However, the complexity and the fine-scale details in the phase space orbits for the high resolution methods indicate that it is very difficult to find an equation describing the behavior. Also, the experiments above show that there is no clear relation between the inner part of the phase space profiles and the amount of post shock oscillations.

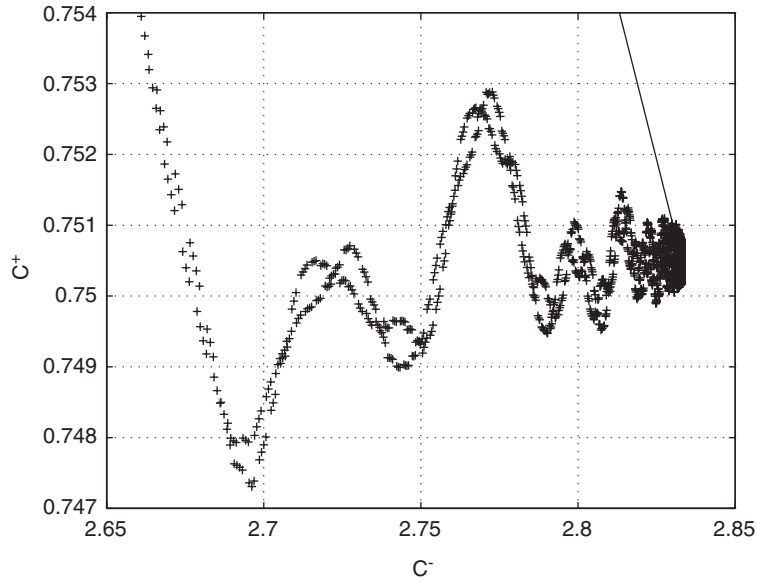


Fig. 15. Zoom of Fig. 14, the phase space plot for the STGU solution of test case T3 with $\Delta x = 0.0025$ and $\Delta t/\Delta x = 0.2$.

2.2. High resolution central schemes

In [26], Nessyahu and Tadmor suggested a family of high resolution central schemes. The schemes can be viewed as extensions of the Lax–Friedrichs scheme and MUSCL reconstruction is used to obtain second order accuracy. We have implemented some of the methods, and, using the same notation as in [26], we denote them LxF, ORD, STG, STG2, STG3 and STGU. Here, LxF is the standard Lax–Friedrichs scheme. ORD is a variant of the scheme on a standard Cartesian grid. In the reconstruction, the Minmod limiter is used. For future reference, we give the definition of the Minmod function:

$$\text{Minmod}(x, y) = \frac{1}{2}(\text{sgn}(x) + \text{sgn}(y)) \min\{|x|, |y|\}.$$

The Minmod function can be similarly extended to include more than two variables. The ORD method involves terms of the type

$$\text{Minmod}(\Delta v_{j+1/2}, \Delta v_{j-1/2}). \quad (4)$$

Here $\Delta v_{j+1/2} = v_{j+1} - v_j$. STG is the scheme corresponding to ORD, but formulated on a staggered grid. A family of variants of the Minmod limiter, parameterized by the parameter α for $\alpha \geq 1$, is introduced in [26, p. 416]. For $\alpha = 1$, the limiter reduces to the standard Minmod limiter defined above. For $\alpha > 1$, this limiter allows steeper slopes in the shock layer. By using this generalized Minmod limiter with $\alpha = 2$ and $\alpha = 3$, respectively, the methods STG2 and STG3 are obtained. Hence, instead of (4), we then have

$$\text{Minmod}(\alpha \Delta v_{j+1/2}, \frac{1}{2}(v_{j+1} - v_{j-1}), \alpha \Delta v_{j-1/2}).$$

STGU is a variant STG which is inspired by the UNO scheme and ENO type methods. In STGU the UNO limiter [14]

$$\text{Minmod}(\Delta v_{j-1/2} + \frac{1}{2}\text{Minmod}(\Delta^2 v_{j-1}, \Delta^2 v_j), \Delta v_{j+1/2} - \frac{1}{2}\text{Minmod}(\Delta^2 v_j, \Delta^2 v_{j+1})) \quad (5)$$

is used. Here, $\Delta^2 v_j \equiv v_{j+1} - 2v_j + v_{j-1}$.

The result from the numerical experiments using this group of high resolution central schemes is in most cases very similar to the result for the Clawpack methods. The numerical profiles obtained seem to indicate the existence of continuum shock profile for the methods. Again, one can see an ordering of the shock profile as the sharpness the limiter allows is increased. Also for these methods, the phase space trajectories are in general very complicated with a

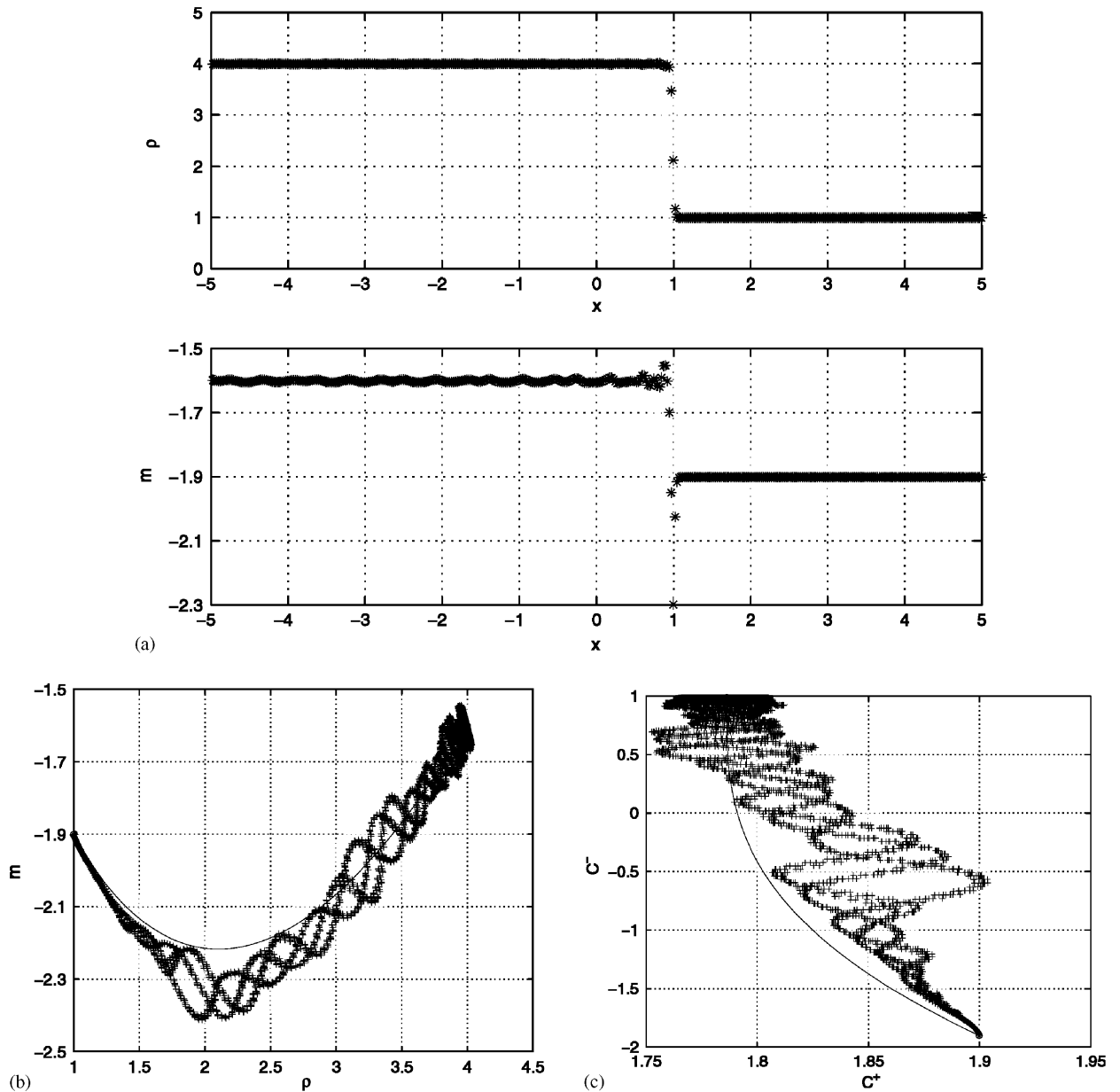


Fig. 16. Solution of test case T1 obtained by the 3-3-LF-ENO with $\Delta x = 0.025$ and $\Delta t/\Delta x = 0.2$: (a) solution in physical space; (b) phase space orbit, ρ - m plane; (c) phase space orbit, C^- - C^+ plane.

lot of sharp corners and fine scale details, and to find an equation describing this behavior seems hopeless. For typical phase space orbits, see Figs. 12 and 13, where phase space orbits obtained by STG and STG2 for test case T1 are shown. In the computations with the high resolution central schemes, we have used the same space and time steps as in the corresponding computation using the CLAWPACK methods. Hence, $\Delta x = 0.0025$ and $\Delta t/\Delta x = 0.32$ for test case T1.

However, for one of the methods, STGU, the result is different. At a first glance, the phase space orbit seems similar to the orbits for the other methods, see Fig. 14, where the phase space orbit obtained by STGU for test case T3 is shown. In the computations, $\Delta x = 0.0025$ and $\Delta t/\Delta x = 0.2$. However, zooming in on the numerical phase space orbit, see Fig. 15, one finds two different orbits. Some of the data points follow one of the orbits, and other points follow the

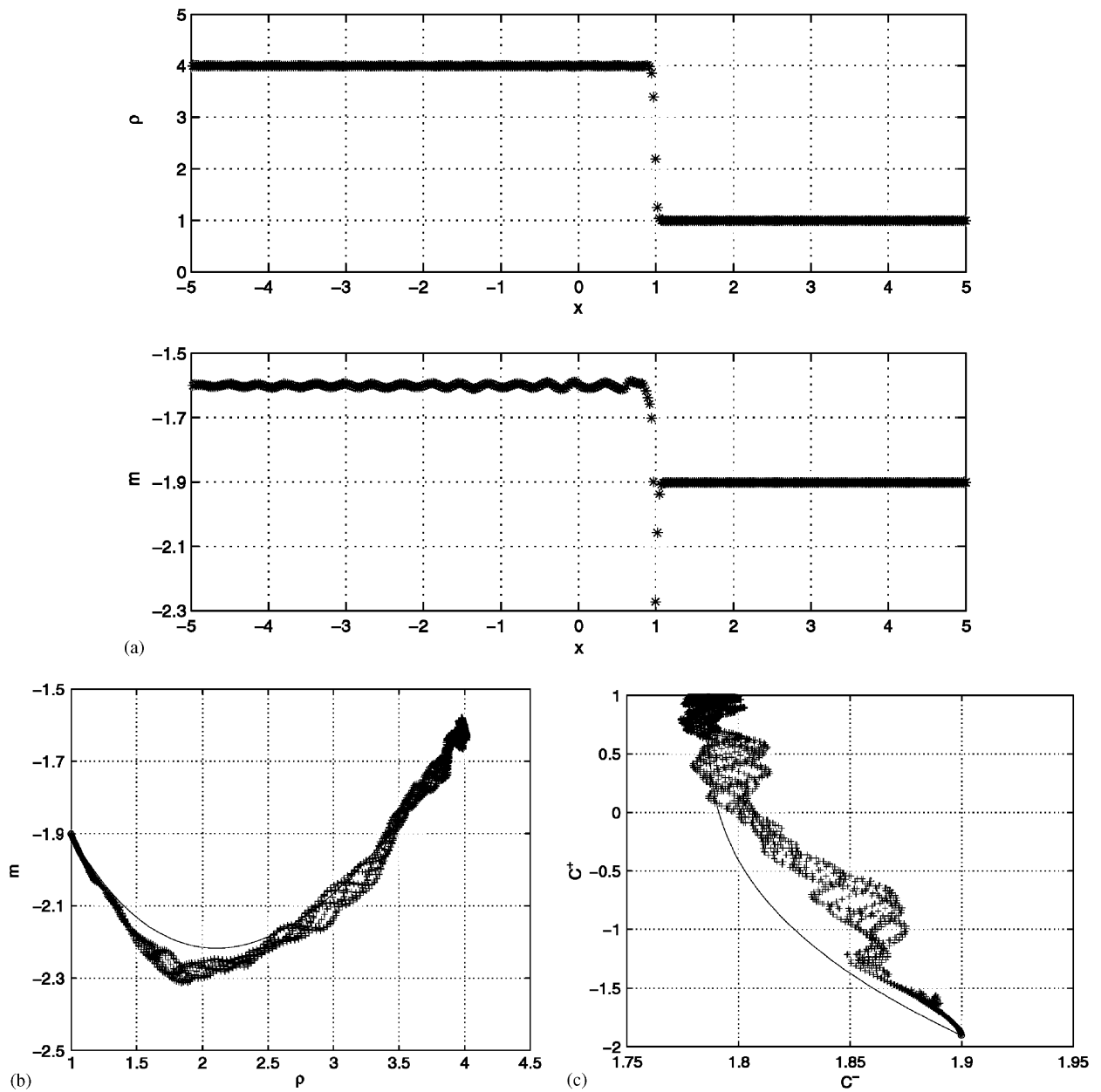


Fig. 17. Solution of test case T1 obtained by the 2-2-LF-ENO with $\Delta x = 0.025$ and $\Delta t/\Delta x = 0.2$: (a) solution in physical space; (b) phase space orbit, ρ - m plane; (c) phase space orbit, C^- - C^+ plane.

other. The two orbits are close to each other. This behavior differs significantly from the result obtained by the other methods investigated so far.

2.3. ENO schemes

The first ENO method was suggested in [15]. The ENO strategy can be used to construct high resolution methods of desired order, and the methods involve piecewise polynomial reconstruction. The stencil for the polynomial reconstruction is chosen adaptively, to obtain an interpolant with as little oscillations as possible. In the original paper [15], the reconstruction procedure was formulated using the Minmod limiter. Therefore, the second order variant of

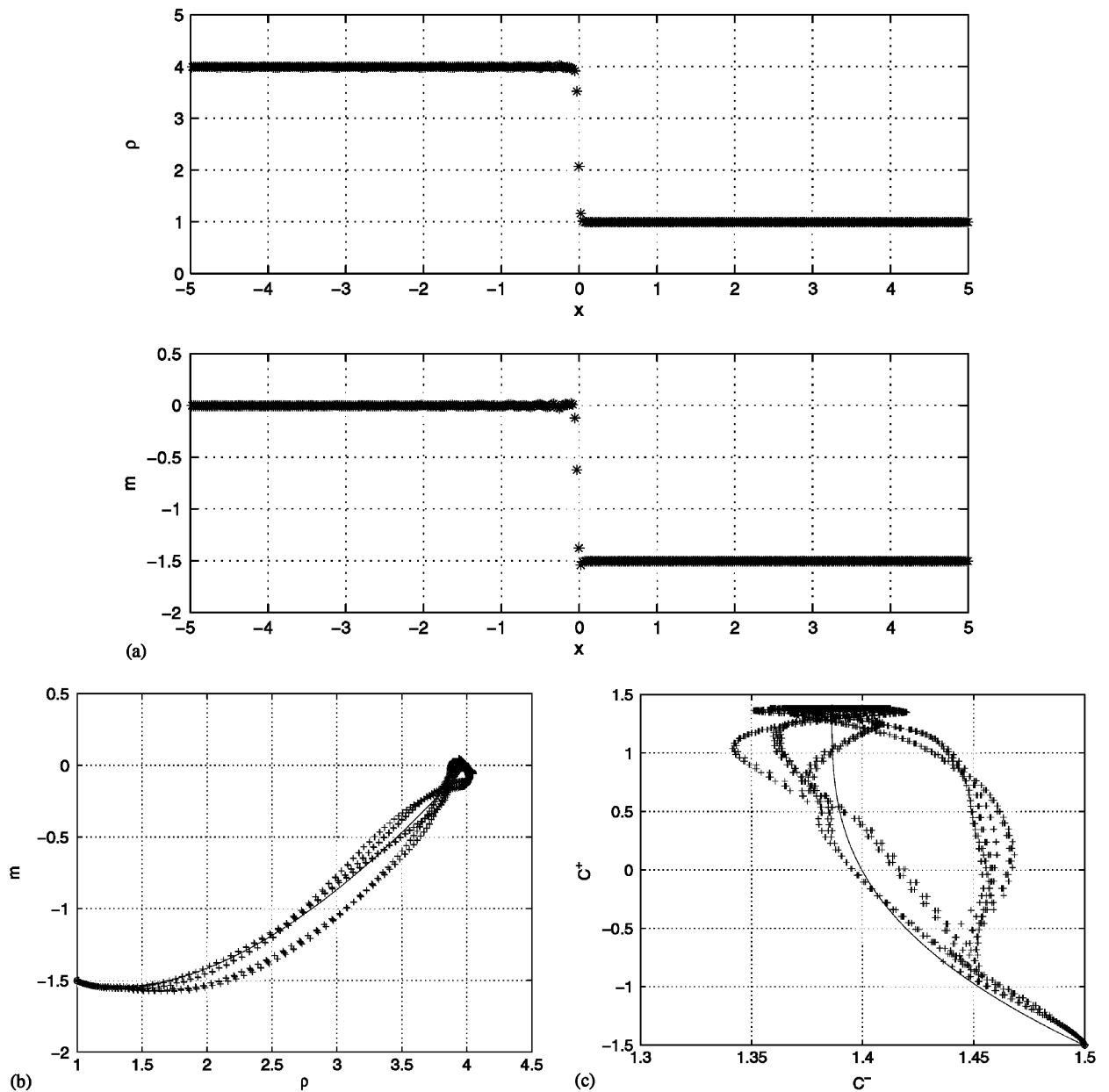


Fig. 18. Solution of test case T2 obtained by the 3-3-LF-ENO with $\Delta x = 0.025$ and $\Delta t/\Delta x = 0.16$: (a) solution in physical space; (b) phase space orbit, ρ - m plane; (c) phase space orbit, C^- - C^+ plane.

the ENO methods suggested in [15] will involve terms similar to (5). Higher order variants of the scheme will involve corresponding terms which include higher order divided differences. In later papers, ENO methods are usually not formulated using limiters in this explicit way.

We have implemented the 2-2-LF-ENO method and the 3-3-LF-ENO method suggested in [29]. In this family of finite difference ENO methods, the space discretization consists of an ENO interpolation applied to the flux, with the Lax–Friedrichs method as building block. Runge–Kutta methods are used for time discretization. The 2-2-LF-ENO is second order accurate in both space and time. The 3-3-LF-ENO is third order accurate in both space and time. In the methods, a constant matrix α must be chosen, and two variants are suggested, see [29, p. 453]. We have used the variant

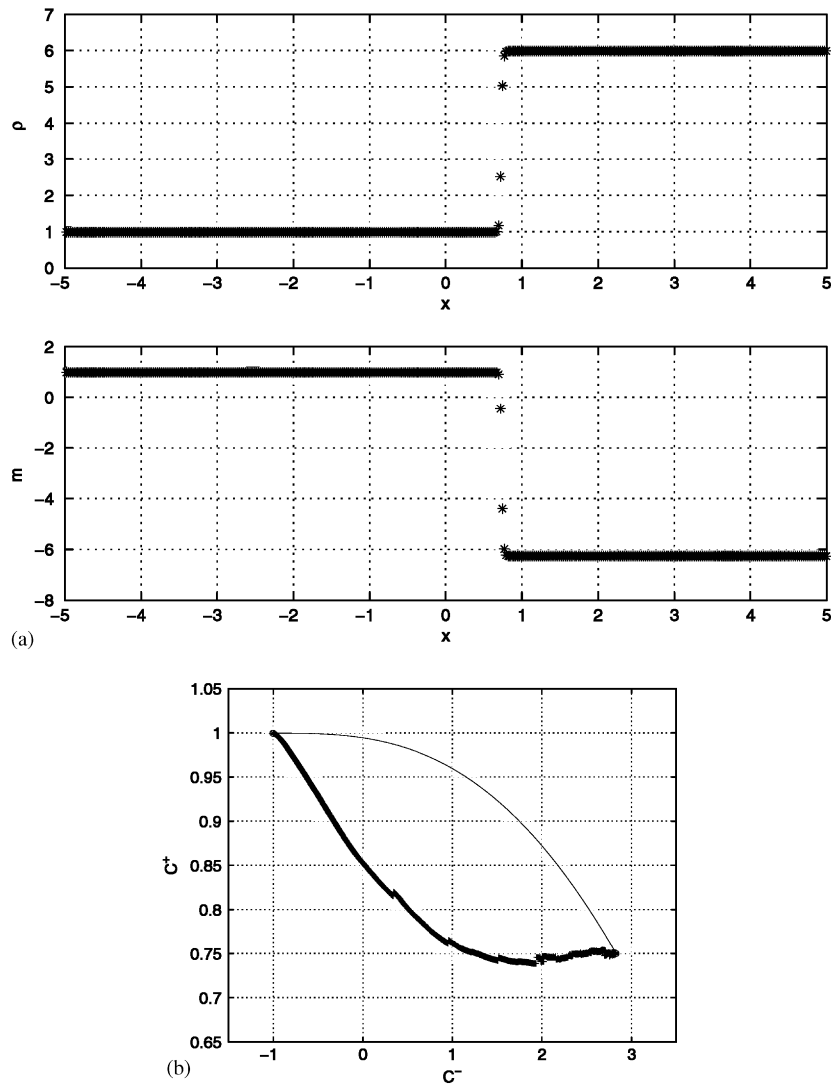


Fig. 19. Solution of test case T3 obtained by the 3-3-LF-ENO with $\Delta x = 0.025$ and $\Delta t/\Delta x = 0.2$: (a) solution in physical space; (b) phase space orbit, C^- – C^+ plane.

$\alpha = \alpha_c I$ where α_c is a scalar constant, which in each computation should be chosen such that $\alpha_c > |\lambda_j(q)|$, $j = 1, \dots, n$. In all computations presented below $\alpha_c = 3$.

The numerical phase space orbits we obtain with the 2-2-LF-ENO method and the 3-3-LF-ENO method differ significantly from the phase space orbits obtained with previous methods, except STGU. Except for STGU, all previous experiments indicate the existence of a continuum shock profile associated with the numerical scheme. However, this is not the case for the 2-2-LF-ENO method and the 3-3-LF-ENO method.

Let us first consider the 3-3-LF-ENO solution of test case T1. The solution is a slowly moving shock. The solution obtained by the 3-3-LF-ENO method is shown in Fig. 16. In the computation, $\Delta x = 0.025$, $\Delta t/\Delta x = 0.2$. When plotted in physical space, as expected, a spike in the momentum variable is seen. In the momentum variable, there are oscillations. In the density variable, the oscillations are much less visible. Let us now study Fig. 16(b), where the phase space orbit for the shock profile is plotted in conserved variables. Instead of one single curve, it here seems to consist of three separate curves. Since T1 has a slowly moving shock, it is easy to follow how each single point travels in phase space, by adding one time step to the phase space plot at a time. Then it can be seen clearly how the points are distributed

among the three curves. Every third points follows the first curve, every third points follows the second curve and finally every third point follows the third curve. Hence, there is not one curve that all discrete data points follow. Also, a closer study shows that there is not exactly three curves. Instead, there is a range of curves that the data points follow.

Our experiments show that there will be multiple phase space orbits also when the order or the ENO interpolation is decreased. However, the spreading of the orbits decreases as the order of interpolation decreases. In Fig. 17, the 2-2-LF-ENO solution for T1 is shown. Again, $\Delta x = 0.025$ and $\Delta t / \Delta x = 0.2$. Here the spreading in phase space of the solution points is smaller.

The character of the phase space orbits varies considerably depending on the specific problem. The numerical solution of T2 has multiple curves in phase space, but the phase space curves do not show the oscillatory pattern that occurred for T1, see Fig. 18. In the computation, $\Delta x = 0.025$, $\Delta t / \Delta x = 0.16$. For test case T3, the phase space orbit seems almost to follow a single curve, see Fig. 19, where $\Delta x = 0.025$, $\Delta t / \Delta x = 0.2$. In all these cases the number of curves changes as $\Delta t / \Delta x$ is varied.

The conclusion of this numerical study is that solutions obtained by the 2-2-LF-ENO method and the 3-3-LF-ENO cannot in general be modeled by a continuum shock profile.

3. Conclusions

We have numerically investigated the phase space orbits for a range of high resolution shock capturing schemes. The aim of the study was to get a better understanding of how these high resolution schemes behave in the shock region, and if possible, to obtain continuous models for the discrete schemes. The first question to investigate is if it is reasonable to model the discrete numerical solution by a continuous function. We have found that it is not always the case. In some cases, the sequence of discrete solutions does not converge to a continuum shock profile, though it remains bounded. Instead, the sequence of discrete solutions tends towards a stable limit cycle. The possibility of such a behavior was anticipated in [6, p. 2285], but no specific examples, or other evidence, of this behavior have previously been given.

For the two ENO methods we have investigated, our numerical experiments show that no continuum shock profiles associated to the schemes exist. Each point in the discrete solution seems to follow a well defined curve in the phase space as the shock traverses it, but different points follow different curves. We have also studied a range of Riemann based flux-limiter methods, and in this case our experiments indicate the existence of associated continuum shock profiles. However, the behavior of the flux-limiter methods in the shock region is very complicated, and it seems impracticable to obtain an equation describing their behavior. For the high resolution central schemes that we have investigated, our experiments indicate the existence of associated continuum shock profiles, for all but one of the investigated schemes, the scheme we denote STGU.

The ENO schemes and the STGU scheme differ from the other investigated schemes. The ENO schemes and the STGU scheme involve terms containing recursive limiting of higher order divided differences of the solution, i.e., terms similar to (5). In the other investigated schemes, limiters are applied to the first divided difference of the discrete solution. We conclude that the existence of a discrete shock profile is heavily dependent on the limiting mechanism.

Other authors have suggested numerical phase space orbits as a tool for investigating oscillations of slowly moving shock. Our study shows that two methods can have very similar phase space orbits, and still have a completely different behavior regarding post shock oscillations.

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